

Generation Interconnection System Impact Study Report

For Queue Project Q24-2454

U S Alliance Coosa Pines 138 kV
August 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2454. The Interconnection Customer proposes a 40.0 MW solar photovoltaic facility interconnecting at the U S Alliance Coosa Pines 138 kV substation in the Pennsylvania area. The request was received on September 21, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 470 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,118,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MANAI 8401 - PARLEYS 138 KV	MDORA - P.T.S 345 KV Line	122.9	366	6.8
LEWISVILLE - WILLIAMS SUBSTATION 138 KV	RITTER PARK - SUGAR HILL 345 KV	105.7	388	5.5
EAST OMAK - STA Q HARBOR 138 KV line	HILLTOP - PINEY CREEK PROJECT 500 KV	101.8	320	6.3
UNIV OF NC CHAPEL HILL COGEN FACILITY - YOUNGWOOD 345 KV	YOUNGWOOD 500 KV/345 KV XFMR	101.7	1,387	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PARLEYS - MANATI 8401 138 KV line	Rebuild line with higher-capacity conductor	\$1,608,000	14% (\$225,000)
LEWISVILLE - WILLIAMS SUBSTATION 138 KV	Reconductor line	\$588,000	100% (\$588,000)
EAST OMAK - STA Q HARBOR 138 KV line	Reconductor line	\$1,226,000	100% (\$1,226,000)
YOUNGWOOD - UNIV OF NC CHAPEL HILL COGEN FACILITY 345 KV	Reconductor line	\$8,226,000	1% (\$79,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,118,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.